

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0268

Jacks Creek 138 kV
September 29, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0268. The Interconnection Customer proposes a 219.3 MW solar photovoltaic facility interconnecting at the Jacks Creek 138 kV substation in the New Jersey area. The request was received on September 01, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 194 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,464,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Jacks Creek-Ardmore 138 KV Line, the Domta 2-Waldwick 138 KV loads to 121.8% of its 248 MVA rating. Project DFAX: 29.3%.

Jacks Creek-Central Plains Wind Farm 138 KV: 119.1% loading (245 MVA) following the outage of the Jacks Creek-Ardmore 138 KV; DFAX = 29.3%.

The Gastonia PrimePower Park-Salton Sea 2 138 KV is overloaded to 112.6% of its 227 MVA emergency rating for the loss of the Homestead-Chickasaw 161 KV. The project contributes 21.7% DFAX (47.7 MW) to this overload.

Waldwick-Central Plains Wind Farm 138 KV: 110.6% loading (228 MVA) following the outage of the Jacks Creek-Ardmore 138 KV L; DFAX = 29.3%.

E 12Th St-Salton Sea 2 138 KV: 108.1% loading (206 MVA) following the outage of the Chickasaw-Homestead 161 KV; DFAX = 21.7%.

Ardmore-Jacks Creek 138 KV: 106.3% loading (246 MVA) following the outage of the Domtar 2-Waldwick 138 KV line; DFAX = 23.6%.

Cuba City S-Deepecreek 161 KV: 100.6% loading (278 MVA) following the outage of the Selkirk Cogen 500-1 kV Transformer; DFAX = 12.3%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Domta 2-Waldwick 138 KV	Rebuild line with higher-capacity conductor	\$2,169,000	77% (\$1,676,000)
Jacks Creek-Central Plains Wind Farm 138 KV	Rebuild line with higher-capacity conductor	\$2,848,000	77% (\$2,200,000)
Gastonia Prime Power Park-Salton Sea 2 138 KV	Rebuild line with higher-capacity conductor	\$445,000	70% (\$312,000)
Waldwick-Central Plains Wind Farm 138 KV	Rebuild line with higher-capacity conductor	\$1,981,000	77% (\$1,530,000)
E 12Th St-Salton Sea 2 138 KV	Reconductor line	\$404,000	70% (\$283,000)
Ardmore-Jacks Creek 138 KV	Reconductor line	\$1,355,000	77% (\$1,047,000)
Cuba City S-Deepcreek 161 KV	Reconductor line	\$912,000	46% (\$416,000)
	Total allocated to project		\$7,464,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.